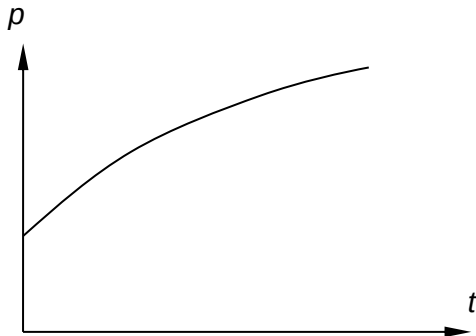


off with same speed as colliding mass.

4.

Ans: C



1. “constant acceleration” $\Rightarrow$ constant $\frac{\Delta v}{\Delta t}$
means constant increase in v
2. but “burns fuel” means mass is decreasing at the same time
3. $\mathbf{p = mv}$, although there is a constant increase in v , there is also a decrease in mass at the same time so it will be a curve rather than straight line graph.

5.

Ans: D

Let l be the distance from hip joint pivot to centre of mass